

## Quiz 2

● Graded

Student

Samyak Sanghvi

Total Points

2.5 / 4 pts

Question 1

(no title)

■ 0.5 / 2 pts

✓ + 0.5 pts Mentioned "Not possible"

+ 1.5 pts Valid counter example

+ 0 pts No/Wrong answer

💬 Reason given is invalid

Question 2

(no title)

2 / 2 pts

✓ + 2 pts Correct explanation

+ 0 pts No/Wrong answer

Name: SamyakCOL7(6)83 Quiz 2  
14 August, 2025Entry no.: 2023CS10807

1. If you only know the unnormalized histograms  $h_f$  and  $h_g$  of two images  $f$  and  $g$ , is it possible to determine the histogram of  $f + g$ ? Justify your answer.

No,

Since histograms involving rounding of intensities  
this may cause an underflow in  $h_{f+g}$   
as we can't directly add  $h_f, h_g$

2. A high-pass filter should output zero in areas of constant intensity. Based on this, explain briefly why the entries of the corresponding kernel  $w$  must sum to zero.

$$w * f(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b f(x-s, y-t) w(s, t)$$

Consider a neighbourhood of  $B \cdot M \times N$  pixels of  
constant intensity pixels  $(2a+1, 2b+1)$

$$\Rightarrow f(x-s, y-t) = c \quad (\text{say}) \quad (\text{in this neighbourhood})$$

$s: -a \rightarrow a$   
 $t: -b \rightarrow b$

$$\therefore w * f(x, y) = 0 = c \cdot \sum \sum w(s, t)$$

Hence sum of entries of kernel is 0 for high pass filter.